

Sort and Bundle Market Produce SOP

One episode clears a tray of mixed produce into two woven baskets, one basket per type, with the large pieces at the back of the basket and the small pieces at the front. It then bundles a loose pile of greens with an elastic band and puts the finished bundle on the scale for one reading.

The order never changes: sort every apple, then sort every potato, then bundle the greens, then weigh the bundle. Inside each basket the large pieces go in first, then the small ones. Nothing is bundled while a piece of produce is still on the produce tray.

The right gripper does the picking, the placing, the bundling, and the weighing. The left gripper holds the basket that is being filled, then holds the greens flat while the band goes on. Neither gripper reaches across the table into the other one's zone.

Every piece is carried one at a time and lowered into the basket until it rests on the basket floor. Nothing is dropped in from above, nothing is dragged across the table, and no gripper ever closes on two pieces at once. The band stand and the scale are never gripped or nudged.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the produce tray at the center front, the apple basket and the potato basket side by side at the back center, the greens board at the front right, the band stand at the right end of the greens board, and the scale at the back right.
3. Both baskets are visible from above, so the back half and the front half of each one can be seen.
4. The band stand is visible from the front, so the open band loop and the stems passing through it can be seen.
5. The scale display is visible and readable in frame.
6. Both arms are at home with grippers open.
7. The table is clean, dry, and bare apart from the produce tray, the two baskets, the greens board, the band stand, and the scale.
8. The left arm reaches the two baskets and the leaf end of the greens board. The right arm reaches the produce tray, the two baskets, the greens board, the band stand, and the scale.

Materials checklist

1. Eight pieces of **produce** lie loose on the produce tray at the center front, mixed rather than grouped: four **apples** and four **potatoes**.
2. Each type has two **large** pieces and two **small** pieces. The difference is obvious. Anyone looking at the tray can tell a large apple from a small one without picking either up.
3. Every piece is firm, dry, and clean, and sits still on the tray instead of rolling.
4. Every piece fits inside one gripper in a single grasp.
5. Two empty **woven baskets** stand side by side at the back center, mouths up. The **apple basket** is on the left and the **potato basket** is on the right. Each one is wide enough to hold four pieces without stacking, and low enough that a piece can be lowered in without lifting it over head height.
6. Each basket is heavy enough to stay put once the left gripper holds its near rim, and light enough that a knock from a gripper would shift it. This is why it gets held.
7. One loose pile of **greens** lies on the **greens board** at the front right, already gathered into a rough bunch, stem ends pointing right toward the band stand and leaves spread to the left.
8. The greens are dry, and the stem ends are level enough to pass through the band loop together.
9. One **band stand** sits at the right end of the greens board: a weighted block with two short upright pegs, one **elastic band** stretched around both pegs so it stands as an open loop above the board surface.
10. The band loop is wide enough for the stem ends to pass through and tight enough to close on the stems once it comes off the pegs.
11. One **scale** sits at the back right, switched on, showing zero, with an empty pan.

Workspace layout

- **Produce tray:** center front, holding the eight mixed pieces.
- **Apple basket:** back center, left of the potato basket, empty.
- **Potato basket:** back center, right of the apple basket, empty.
- **Greens board:** front right, holding the loose greens, stem ends right and leaves left.
- **Band stand:** at the right end of the greens board, carrying the open band loop.
- **Scale:** back right, empty and showing zero.

Everything the right gripper carries stays on the right and the center. The left gripper only ever holds something in place. It never carries anything across the table.

Vocabulary

- **Arm assignment:** the right gripper picks every piece, places it in a basket, slides the greens into the band, lifts the bundle, and works the scale. The left gripper holds the basket

being filled and holds the greens flat on the board. Neither gripper takes the other one's work.

- **Piece:** one apple or one potato. Pieces are picked one at a time and never two in one grasp.
- **Large and small:** the two obvious sizes inside one type. Told apart by eye on the tray, not measured.
- **Basket back:** the half of the basket farthest from the front edge. All large pieces go here.
- **Basket front:** the half of the basket nearest the front edge. All small pieces go here.
- **Rim hold:** the left gripper closed on the near rim of a basket, pressing it down against the table, so the basket cannot slide or tip while the right gripper places a piece. The hold stays on for the whole step.
- **Settled placement:** the piece is resting on the basket floor or on another piece, is not rolling, and is not leaning on the basket wall. The right gripper opens only once this is true.
- **Greens:** the loose pile of leafy stalks on the greens board. The **stem ends** are the cut bottoms pointing right. The **leaf end** is the spread leafy side on the left.
- **Leaf hold:** the left gripper closed on the greens at the leaf end, pressing them flat against the greens board, so the bunch stays gathered while the right gripper slides it. The greens never leave the board during a leaf hold.
- **Open band loop:** the elastic band as it stands stretched around the two pegs of the band stand, with a gap under it the stem ends can pass through.
- **Band set:** the band has come off both pegs and closed around the stems by itself. Seen as the band gripping the stems all the way round with no peg still through it.
- **Bundle:** the greens once the band is set on them, held together as one piece that can be lifted by the stems.
- **Steady reading:** the scale display has stopped changing and shows one number for a count of two, with the bundle resting fully on the pan and nothing else touching it.
- **Lift:** coming straight down onto the thing, closing, and raising straight up until there is daylight under it, before anything moves sideways.
- **Release point:** over a basket, over the greens board, or over the scale pan. A gripper opens nowhere else.

Steps

Steps 1 and 2 empty the produce tray into the two baskets. Step 3 bundles the greens. Step 4 weighs the bundle. Step 5 ends the episode.

Step 1: Sort the apples into the apple basket

Goal: all four apples in the apple basket, the two large ones at the basket back and the two small ones at the basket front, and no apple left on the produce tray.

1.1 Take the rim hold

- The **left gripper** closes on the near rim of the **apple basket** and holds it down against the table.

The rim hold stays on until the end of Step 1.

1.2 Place the two large apples

- The **right gripper** closes on one **large apple** on the produce tray, lifts it straight up, carries it level to the apple basket, lowers it into the **basket back** until it rests on the basket floor, and releases on a **settled placement**.
- Repeat for the second large apple, setting it beside the first rather than on top of it.

Take one apple per grasp. Never drop an apple in from above the rim, never roll or drag one across the table, and never take a potato in this step.

1.3 Place the two small apples

- The **right gripper** places each **small apple** the same way into the **basket front**, one at a time, the two of them side by side.
- The **left gripper** then releases the rim.

Check: four apples sit in the apple basket, both large ones at the basket back and both small ones at the basket front, none stacked on another, no apple outside the basket, the basket has not moved, and only potatoes and the greens are left on the produce tray and the greens board.

Step 2: Sort the potatoes into the potato basket

Goal: all four potatoes in the potato basket, large at the basket back and small at the basket front, and the produce tray empty.

- The **left gripper** closes on the near rim of the **potato basket** and holds it down against the table.
- The **right gripper** places the two **large potatoes** into the **basket back**, one at a time, each on a settled placement.
- The **right gripper** places the two **small potatoes** into the **basket front** the same way.
- The **left gripper** releases the rim.

Large before small, one potato per grasp, and no apple is touched again in this step.

Check: four potatoes sit in the potato basket, both large at the basket back and both small at the basket front, the produce tray is empty, both baskets are still in their places at the back center, and nothing has rolled onto the table.

Step 3: Bundle the greens

Goal: the band set on the stems and the greens held together as one bundle.

3.1 Hold the greens flat

- The **left gripper** closes on the greens at the **leaf end** and presses them flat against the **greens board**, gathering the stalks together.

3.2 Slide the stems into the band

- The **right gripper** closes on the greens across the stalks, between the leaf hold and the stem ends.
- Slide the greens straight right along the board so the **stem ends** pass under the **open band loop**.
- Stop when the band sits across the stalks well above the stem ends, so it cannot slip off the ends.

Slide along the board. Never lift the greens into the air to feed them through, never push the band stand along with them, and never split the bunch into two grasps.

3.3 Set the band and free the bundle

- The **left gripper** releases the leaf end.
- The **right gripper** lifts the greens straight up off the band stand, so the band slides up both pegs and comes off, closing on the stalks by itself.

Check: the band grips the stalks all the way round with no peg through it, the band sits above the stem ends rather than at them, the greens hang together as one **bundle**, no stalk has been left behind on the board, and the band stand is still in its place at the right end of the greens board.

Retry: if the band stays on a peg or slips off the stem ends, the **right gripper** lays the greens back on the greens board with the stem ends pointing right, and Step 3 runs again from 3.1.

Step 4: Weigh the bundle

Goal: one steady reading on the scale with the bundle resting on the pan, then the bundle back on the greens board.

- The **right gripper** carries the **bundle** level to the **scale**, lowers it onto the pan until it rests flat, and releases.
- Wait for a **steady reading**.
- The **right gripper** lifts the bundle straight up off the pan, carries it to the **greens board**, lays it down with the stem ends pointing right, and releases.

Let the pan carry the whole bundle. Never hold the bundle while it sits on the pan, never let a leaf hang off the pan onto the table, and never press down on the pan or nudge the scale.

Check: the display showed one steady reading with the bundle resting fully on the pan and nothing else touching it, the bundle now lies on the greens board with the band still set, and the

scale is back to empty and has not moved.

Step 5: Confirm the result and end the episode

- Confirm the apple basket holds four apples with the large ones at the basket back, the potato basket holds four potatoes the same way, the produce tray is empty, the bundle lies on the greens board with the band set, the band stand is clear, and the scale pan is empty.
- Return both arms home, clear of the baskets and the greens board, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the eight pieces out of the two baskets and lay them back on the produce tray, mixed rather than grouped, four apples and four potatoes, two large and two small of each.
2. Wipe both baskets out, check that neither is holding a loose stalk or a piece of skin, and stand them side by side at the back center, apple basket on the left, mouths up.
3. Slide the band off the bundle and throw it away. A band is used once.
4. Spread the greens back into a loose gathered pile on the greens board at the front right, stem ends pointing right and leaves left.
5. Stretch a fresh elastic band around both pegs of the band stand so it stands as an open loop, and set the stand back at the right end of the greens board.
6. Clear the scale pan, wipe it dry, and confirm the display reads zero.
7. Wipe the table and confirm the surface is clean and dry. A wet table lets the baskets and the greens board slide.
8. Replace a piece of produce that is bruised, soft, wet, or split, greens that are wilted or wet, a band that is slack or torn, and a basket that is split, misshapen, or rocking on the table.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a potato goes into a basket before all four apples are in theirs, the greens are bundled before the produce tray is empty, or the bundle goes on the scale before the band is set. Inside a basket, a small piece goes in before both large ones.
- **SOP rule broken:** Steps 1 to 4, all apples, then all potatoes, then the bundle, then the weigh, with large pieces placed before small ones in each basket.
- **Coaching note:** clear the tray before you touch the greens, and always place large before small.

Violation: Arms swapped roles

- **Visible cue:** the left gripper picks a piece off the produce tray, places one in a basket, slides the greens, lifts the bundle, or touches the scale, or the right gripper takes the rim hold or the leaf hold.
- **SOP rule broken:** Steps 1 to 4, the right gripper picks, places, slides, lifts, and weighs, and the left gripper only holds the basket rim and the leaf end.
- **Coaching note:** the left gripper holds. It never carries.

Violation: Piece in the wrong basket

- **Visible cue:** at the end of Step 2 an apple sits in the potato basket or a potato sits in the apple basket, or a piece is left on the produce tray, on the table, or on the floor.
- **SOP rule broken:** Steps 1 and 2, all four apples into the apple basket and all four potatoes into the potato basket, leaving the produce tray empty.
- **Coaching note:** name the type before you close the gripper, not on the way to the basket.

Violation: Large and small not split back to front

- **Visible cue:** at the end of Step 1 or Step 2 a large piece sits in the basket front, a small piece sits in the basket back, the sizes are mixed through the basket, or a piece is stacked on another instead of set beside it.
- **SOP rule broken:** Steps 1.2, 1.3, and 2, large pieces into the basket back and small pieces into the basket front, none stacked.
- **Coaching note:** back half is for the big ones. Fill it first and the front stays clear for the small ones.

Violation: Piece dropped in, dragged, or doubled up

- **Visible cue:** the right gripper opens above the basket rim and the piece falls in, a piece is rolled or dragged across the table instead of lifted and carried, two pieces are taken in one grasp, or the gripper releases while the piece is still rolling or leaning on the basket wall.

- **SOP rule broken:** Steps 1.2, 1.3, and 2, one piece per grasp, lifted and carried level, lowered until it rests, and released only on a settled placement.
- **Coaching note:** take it down to the floor of the basket, wait until it sits still, then open.

Violation: Basket not held or knocked out of place

- **Visible cue:** the right gripper places a piece while the left gripper is off the rim, the left gripper holds the basket that is not being filled, the rim hold comes off before the last piece of the step is settled, or a basket is struck and slides, turns, or tips.
- **SOP rule broken:** Steps 1.1, 1.3, and 2, the rim hold is on the basket being filled and stays on for the whole step.
- **Coaching note:** hold first, place second. The rim comes free only when the step is done.

Violation: Greens not held flat

- **Visible cue:** the right gripper slides the greens while the left gripper is off the leaf end, the bunch splay apart on the board, stalks are left behind when the greens move, or the greens are lifted into the air to feed them through the band.
- **SOP rule broken:** Steps 3.1 and 3.2, the leaf hold presses the greens flat on the board and the greens slide along the board rather than through the air.
- **Coaching note:** gather them under the left gripper first. Everything in this step happens on the board.

Violation: Band not set on the stems

- **Visible cue:** at the end of Step 3 the band is still on a peg, the band has slipped off the stem ends and stayed on the stand, the band sits down at the stem ends instead of above them, the band has caught the leaves rather than the stalks, or the greens come away loose with no band on them.
- **SOP rule broken:** Step 3, the stem ends pass under the open band loop, the band sits across the stalks well above the stem ends, and the straight lift takes it off both pegs onto the stalks.
- **Coaching note:** push far enough in that the band is on the stalks, then lift straight up. Do not pull sideways.

Violation: Weighing wrong

- **Visible cue:** the bundle is read while the right gripper still holds it, part of the bundle hangs off the pan or rests on the table, the bundle is lifted off before the display stops changing, the pan is pressed down or the scale is nudged, or the bundle is set down anywhere other than the greens board afterwards.
- **SOP rule broken:** Step 4, the bundle rests free on the pan for a steady reading, then goes back on the greens board stem ends right.
- **Coaching note:** let go, wait for the number to settle, then pick it up.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** a piece, the greens, or the bundle falls to the table or the floor short of its spot, the produce tray, the greens board, the band stand, or the scale is struck and shifts, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 4, keep each thing on a clear travel path and release only over a release point.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a piece still on the produce tray, a piece in the wrong basket, the greens loose or still on the band stand, the bundle still on the scale pan, an arm away from home, or an arm still over a basket.
- **SOP rule broken:** Step 5, confirm the two baskets, the empty tray, the bundle on the greens board, and the empty pan; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective produce:** a piece that is soft, split, or wet enough to slip out of a correct grasp, or one that will not sit still on the tray. Replace it before the next episode.
- **Defective greens:** wilted or wet stalks that will not gather under a correct leaf hold, or stem ends so uneven they cannot pass through the loop together. Replace them before the next episode.
- **Band fault:** the band is slack and will not close on the stalks, snaps during the lift, or will not stand as an open loop on the pegs. Replace it before the next episode.
- **Defective basket:** split, misshapen, or rocking on the table, or so light that it lifts with a correct rim hold. Replace it before the next episode.
- **Scale fault:** the display will not settle, will not return to zero, or is unreadable in frame. Replace or rezero it before the next episode.
- **A basket, the greens board, or the band stand slides** under a correct hold because the table is wet or slick. Wipe and dry the table before the next episode.

Annotation subtasks (from SOP)

1. Hold the apple basket rim with the left gripper
2. Place the two large apples in the basket back
3. Place the two small apples in the basket front
4. Hold the potato basket rim with the left gripper
5. Place the two large potatoes in the basket back
6. Place the two small potatoes in the basket front
7. Hold the greens flat on the greens board with the left gripper
8. Slide the stem ends through the open band loop
9. Lift the greens off the band stand to set the band
10. Weigh the bundle on the scale and read it
11. Lay the bundle back on the greens board
12. Confirm the result and end the episode